
TEAM DEKA DESIGN REVIEW

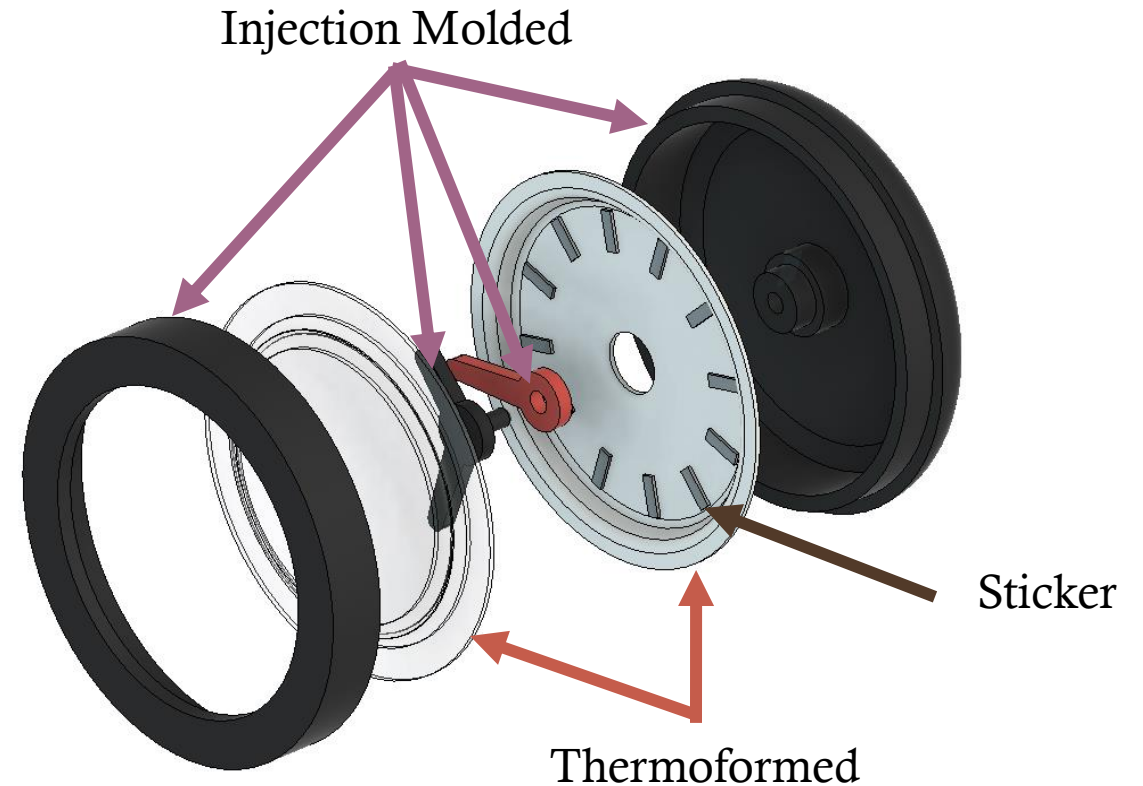
Alex Shanafield, Daniel Gliksberg, Emily Ha, Katherine Jankowski

YOYO THEME - CLOCK



PART AND MOLD COUNT

Injection Molding (IM) Molds	4
Thermoform (TF) Molds	2
IM Parts	4
TF Parts	2
Sticker(s)	1



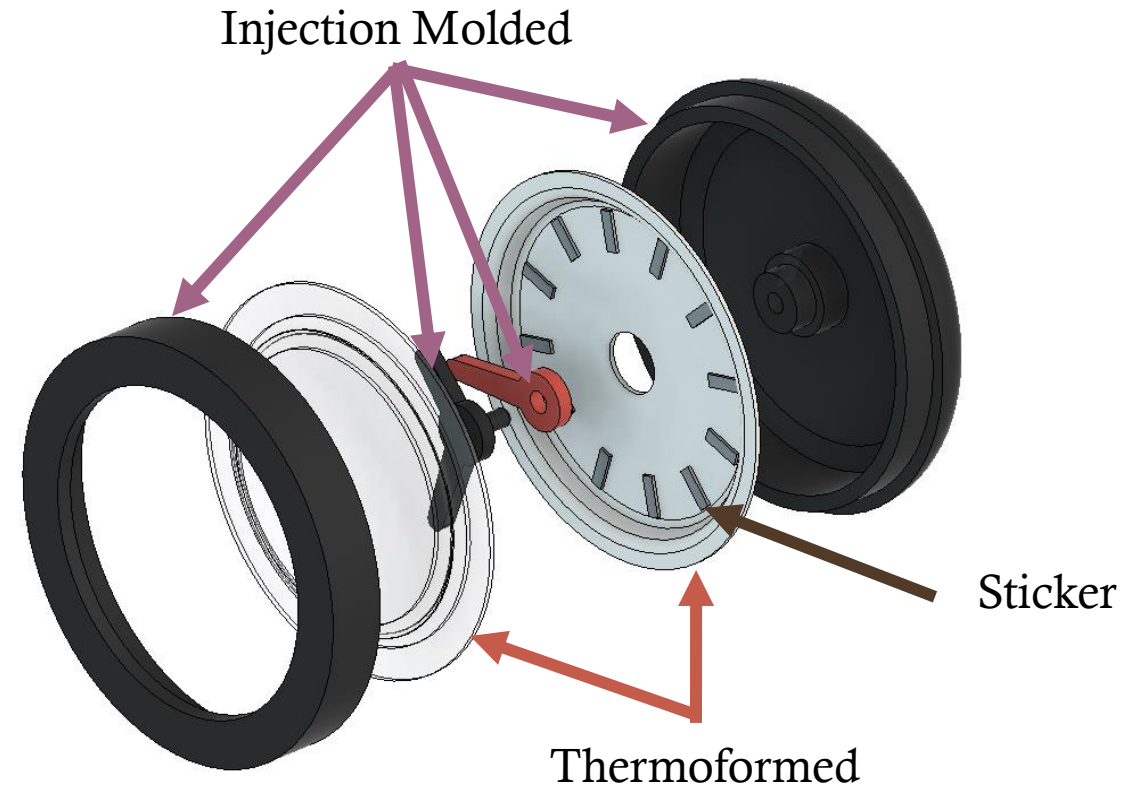
Note: Some colors changed for visibility, overmolded hardware not included

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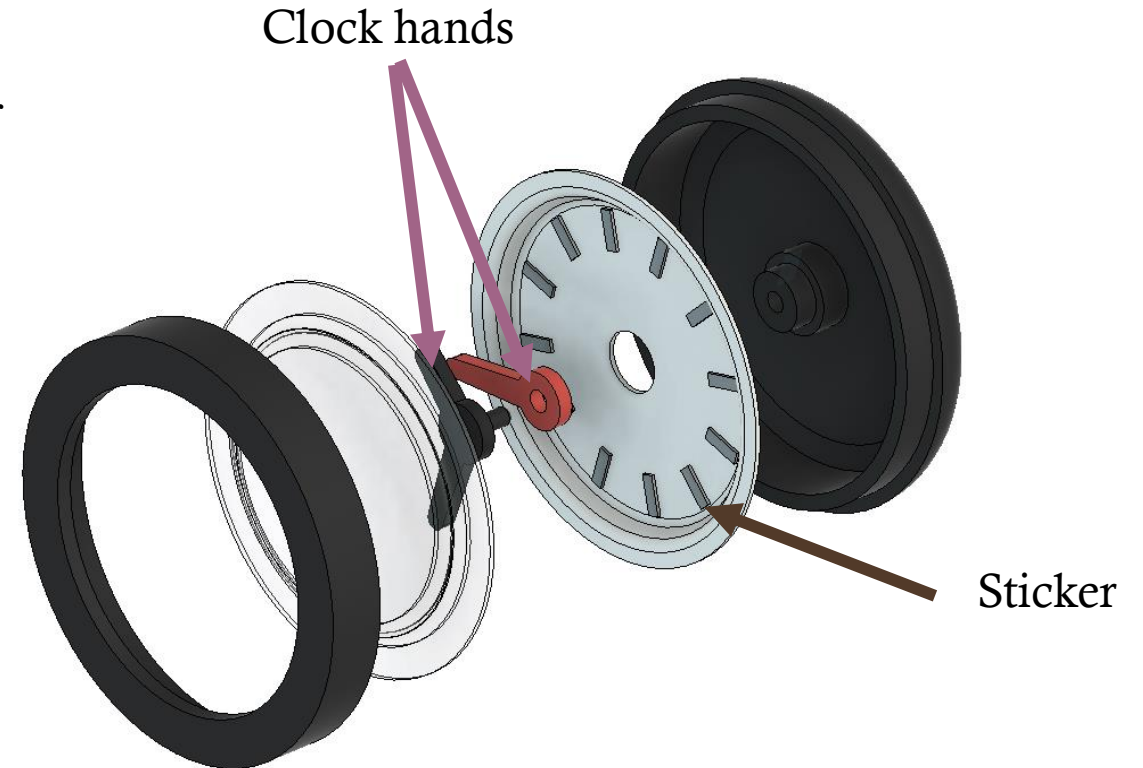


Note: Some colors changed for visibility, overmolded hardware not included

EXPLODED VIEW

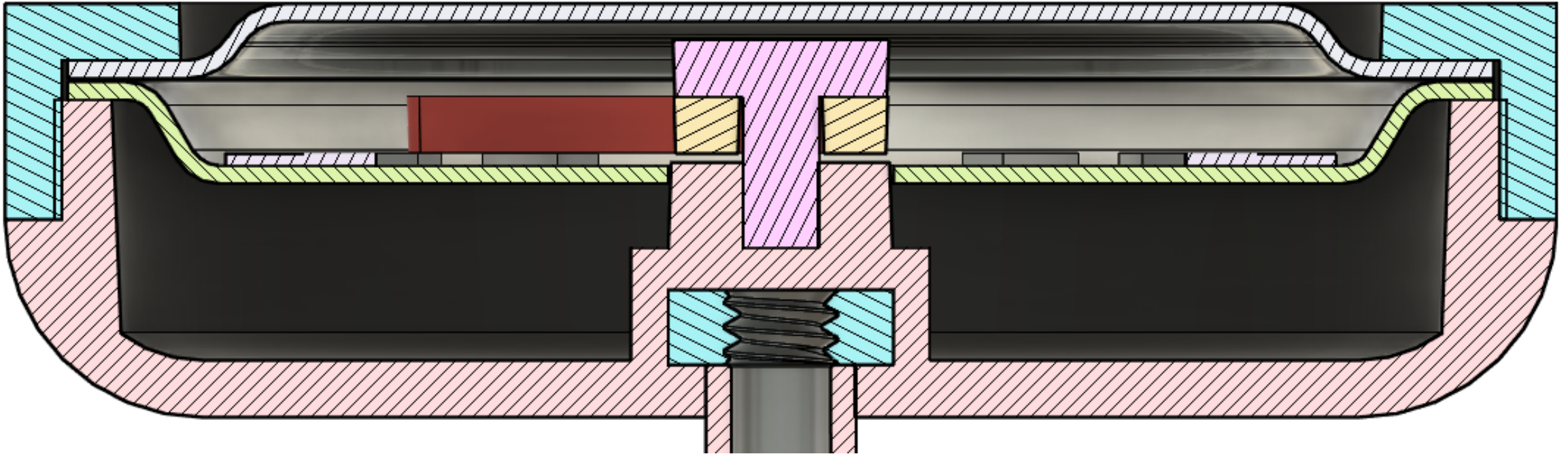
Seconds hand (red) allowed to spin freely
Retained on one side by bottom thermoform, other
side by press-fit hour/minute hand

Sticker applied to bottom TF part before assembly



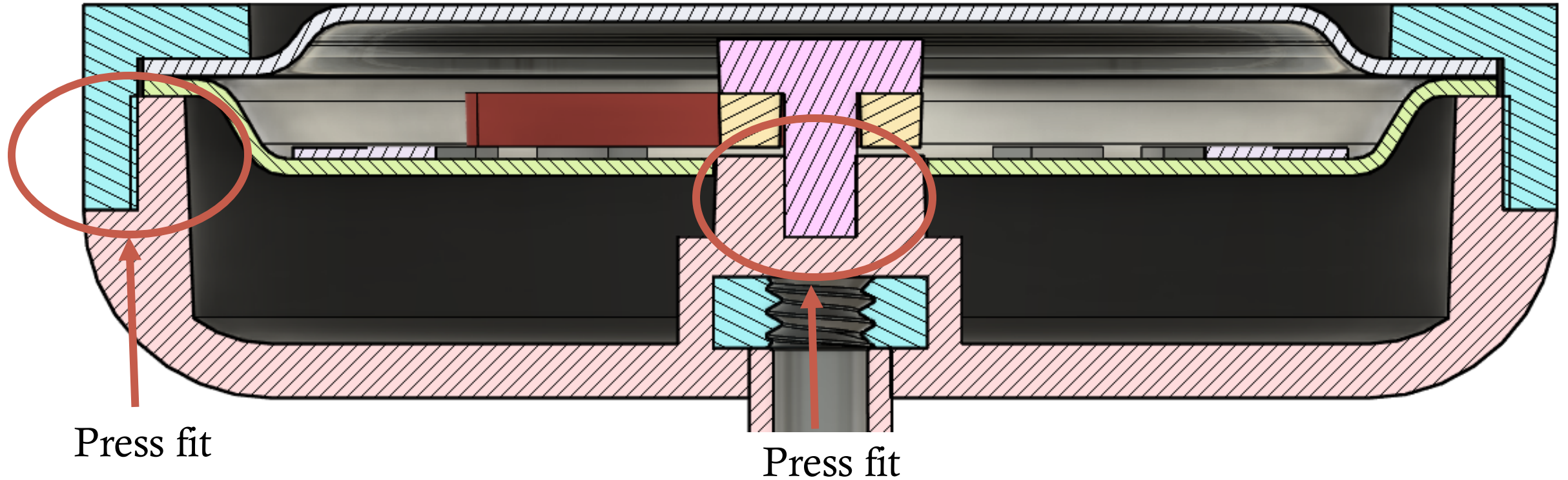
Note: Some colors changed for visibility, overmolded hardware not included

SECTION VIEW



Note: For assembly images, parts are all shown un-scaled to show design intent.

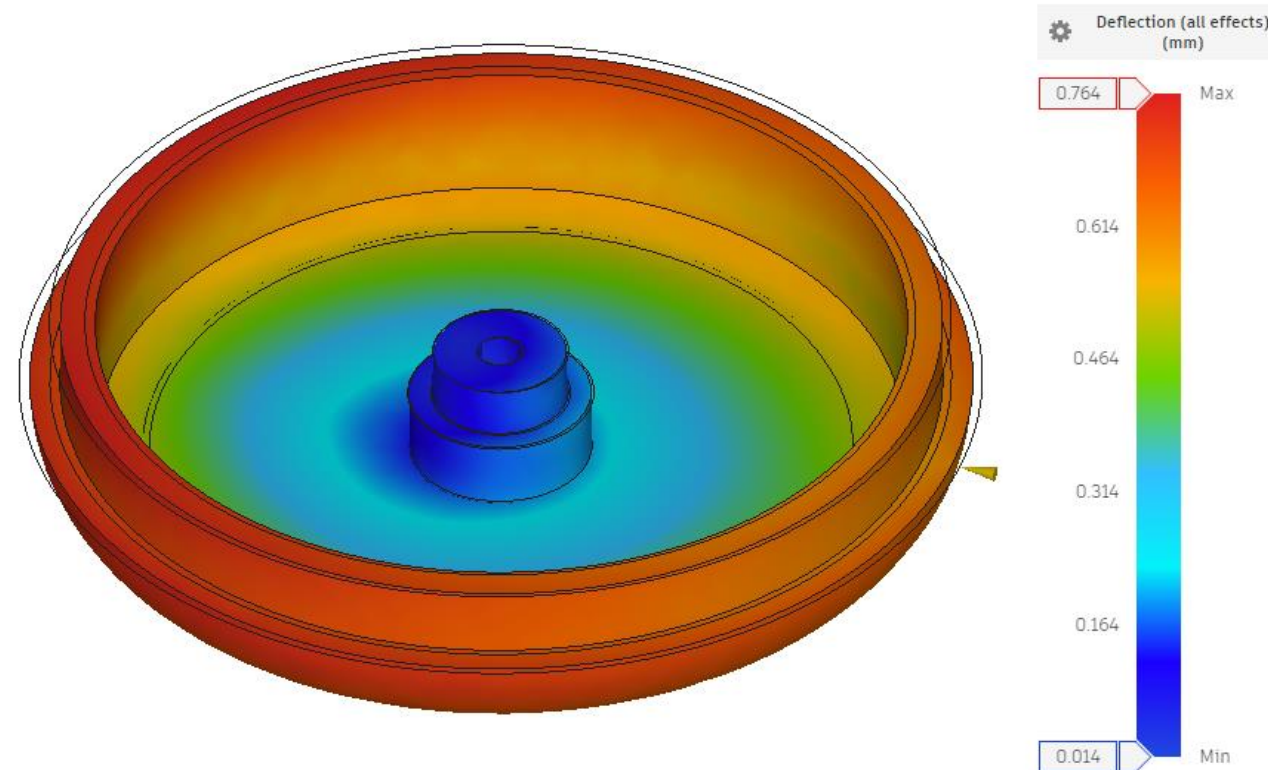
SECTION VIEW



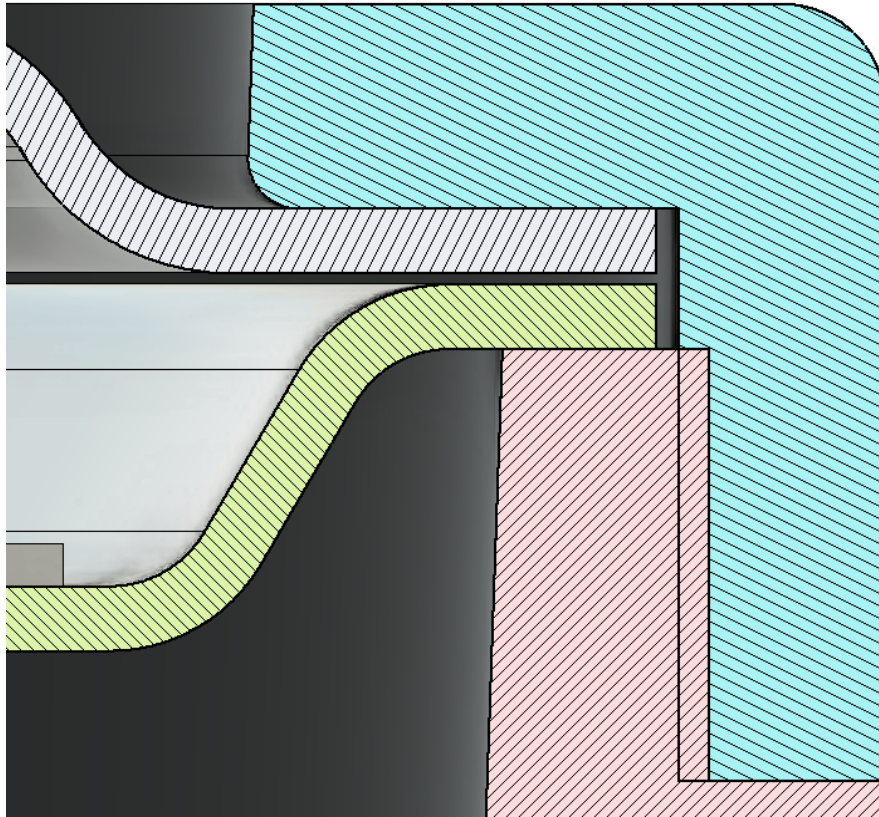
Note: For assembly images, parts are all shown un-scaled to show design intent.

SHRINKAGE MODELING

- Start with MoldFlow estimate
 - Take largest deflection along most critical dimension
 - From there, make a mold and use it to dial in IM parameters
- Make 9x (or nearest multiple if multiple copies per mold), determine error between nom. and mean for most critical dim.
 - Use this to fine-tune % shrinkage, re-machine

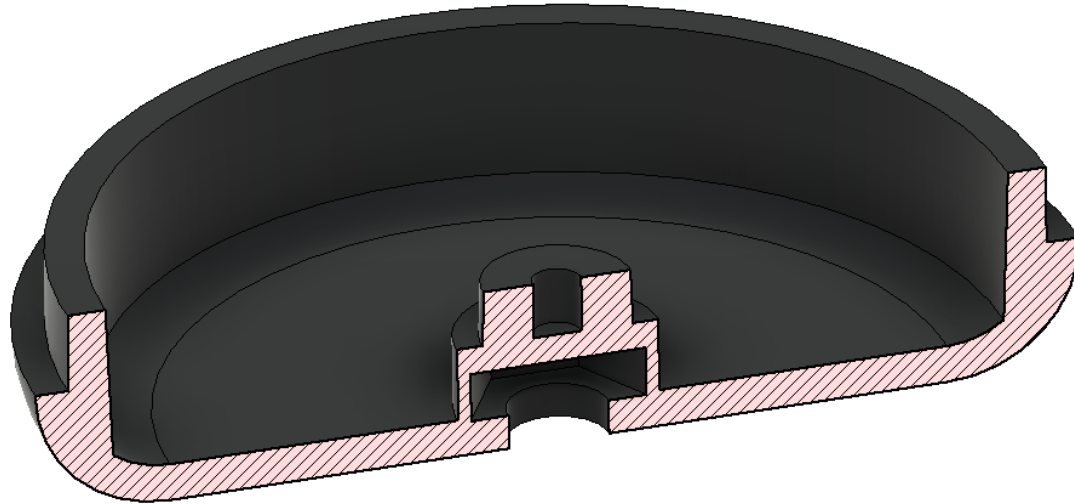


PRESS FIT – YOYO BODY & RING



- Radial interference: 0.012"
- Engagement length: 0.2"
- Determined with CAD Demo video as reference
 - Required separation force calculated from example parameters, then held as constant to determine interference from our parameters
 - $F = 366 \text{ N}$
 - Engagement length is second free variable, kept the same as demo
 - See Appendix 1 for derivation
- Axial & Radial tolerance left for TF parts

YOYO BODY

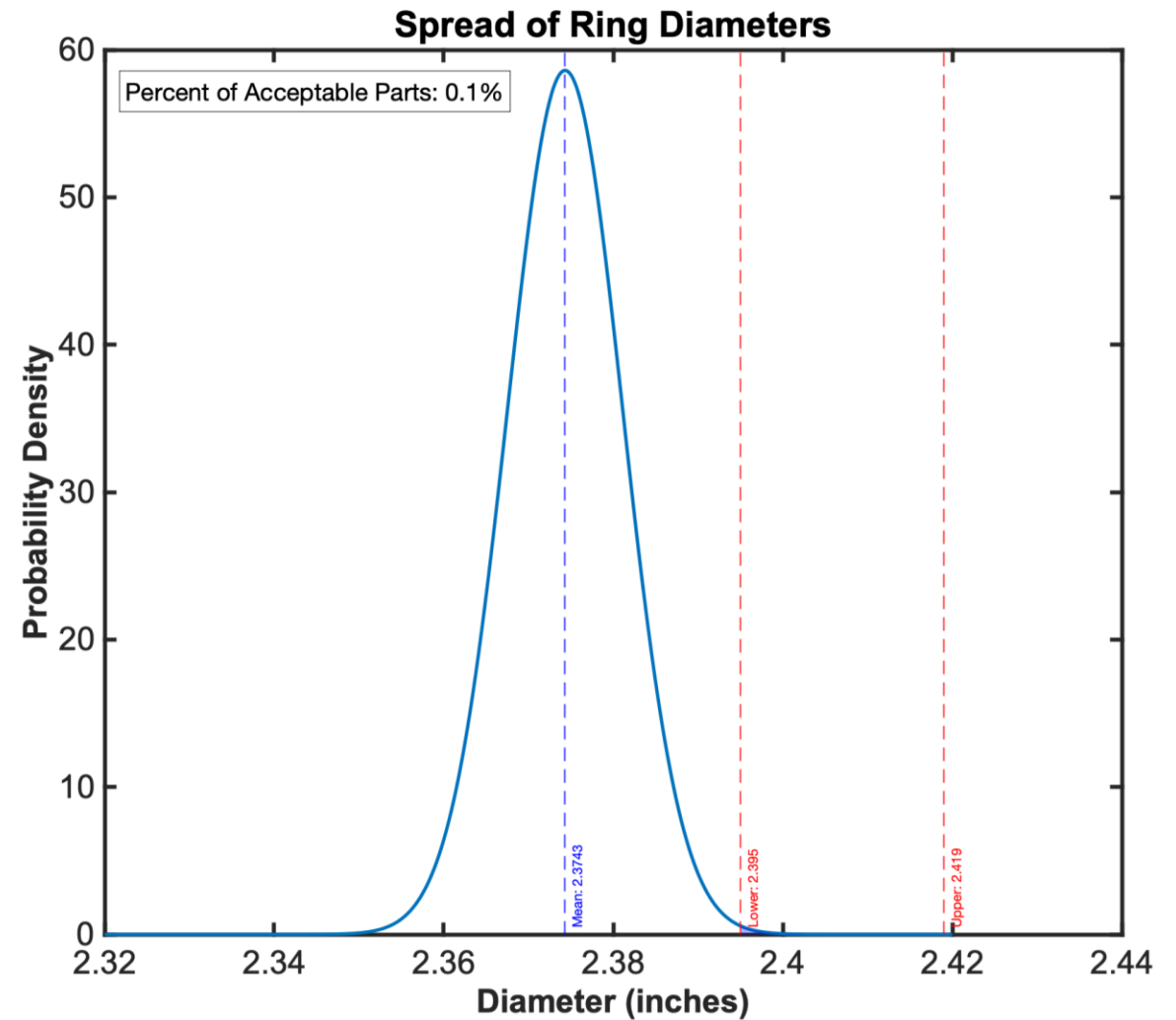


- 0.095" Wall thickness
 - Base of cup, press fit wall
 - Thickens for outer edge, but not enough for sink marks
- Ring above overmolded nut for second press fit
 - Height determined by height of cavity between thermoformed parts
 - More on this press fit in hour-minute hand section
- Draft angles on all internal faces except press fits
 - Not on faces what will shrink away from walls
 - 2° here and everywhere else
- 2.51% Shrinkage around ring press fit from MoldFlow, resulting in a 1.0257 Scaling factor

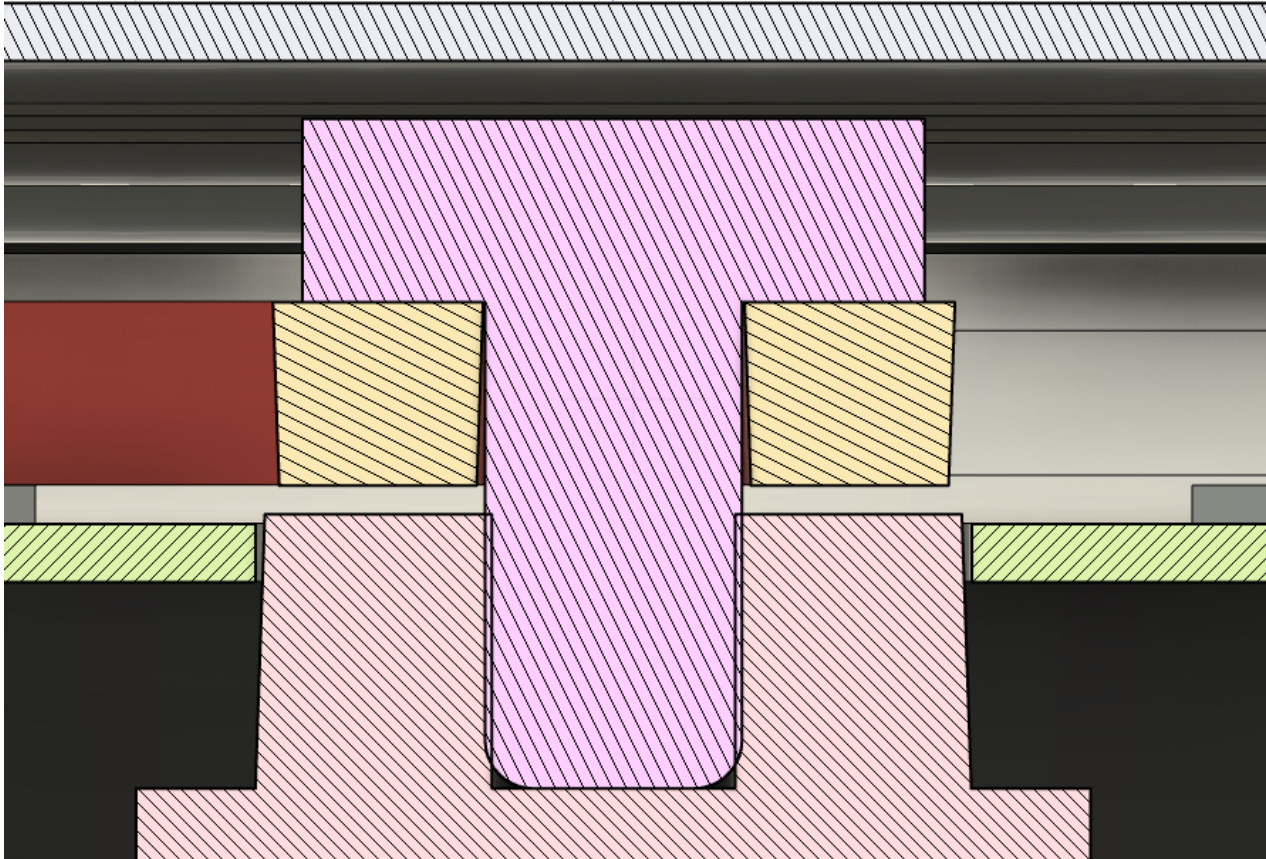
YOYO RING



- 0.095" Wall thickness
- Interference chosen for constant 366 N force
 - 0.012" Radially
- Width of upper lip sized to fit ejector pins
- Draft angle on upper internal radius
 - Only other internal radius is a press fit surface
- 2.8% measured shrinkage: 1.029 Scaling factor
(data from 9 IM rings)

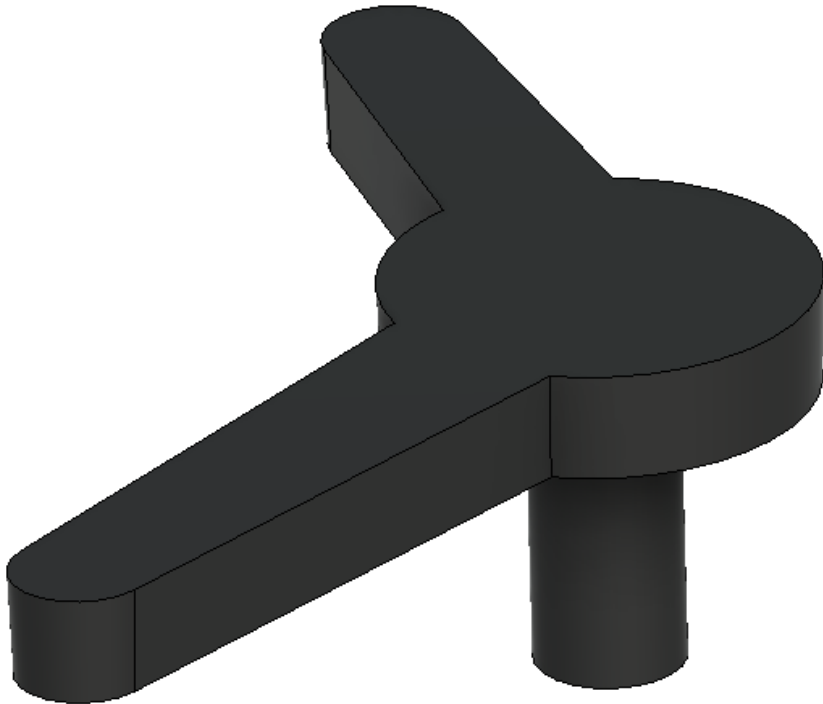


PRESS FIT – YOYO BODY & HOUR/MINUTE HAND



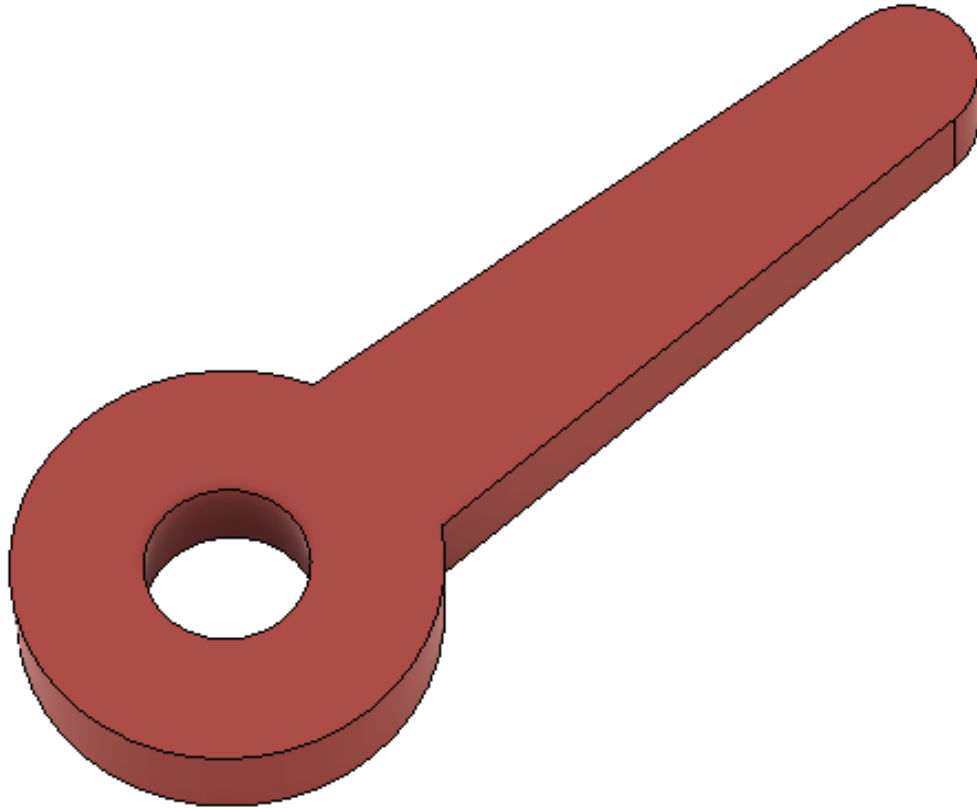
- Radial Interference: 0.002"
- Engagement Length: 0.142"
 - Determined from height of yoyo body and gap between thermoformed parts
- From same equation as before, with same desired separation force
- Note that this interference is a very tight tolerance to hit
- Backup retention method desired
 - Clear thermoform bubble positioned to prevent hour/minute hand from falling out

HOUR – MINUTE HAND



- 0.095" Hand Thickness
- Press fit post is made via intentionally undersized ejector pin
 - Relies on reamed surface finish and either finding or making ejector pin of desired length
- Draft angle on external hand faces
- Will have 8.03% shrinkage on the post, with some sink marks in the center of the larger top circle
 - Marks can be reduced by thinning hands, if we decide we care

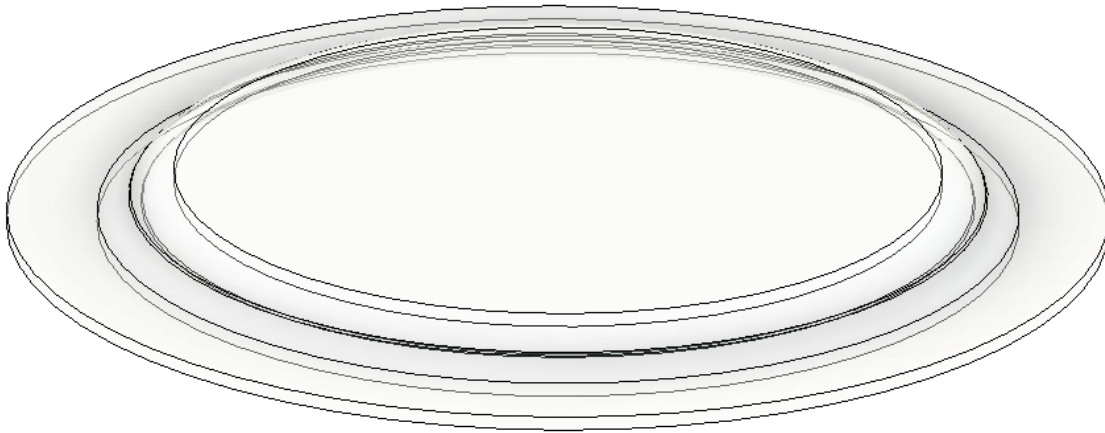
SECONDS HAND



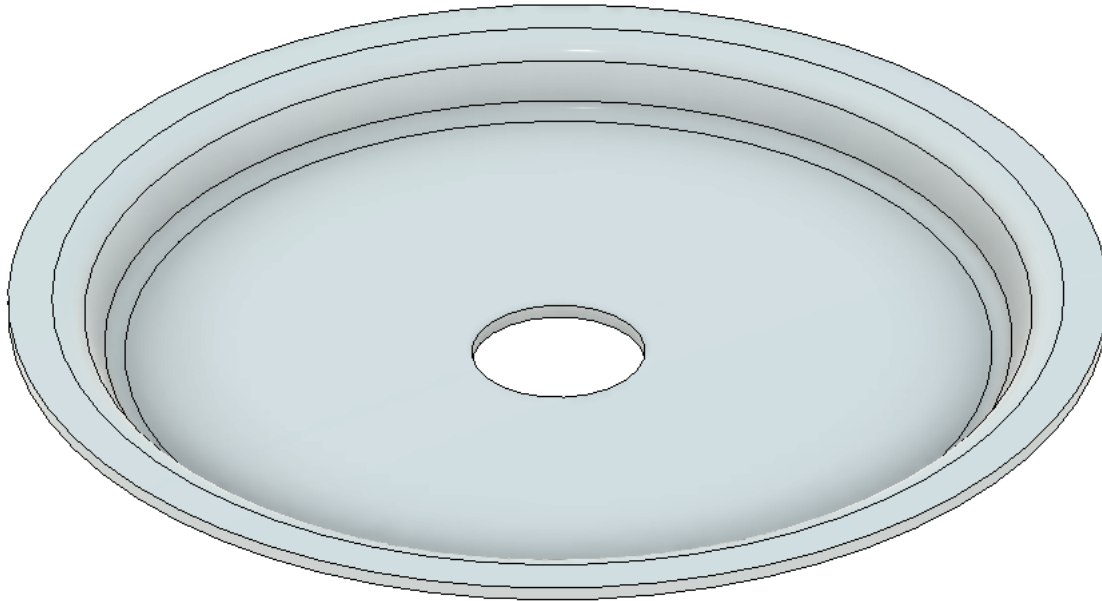
- 0.095" Hand Thickness
 - If hour-minute made thinner, this will match
- No press fits
 - Designed to spin freely, toleranced accordingly
- Draft angle on all vertical faces
- 2.90% Shrinkage determined from Fusion360 Simulations, resulting in 1.0299 Scaling factor

TOP THERMOFORM

- No vacuum holes, test part showed that they are unnecessary
- 2 3/8" die punch diameter
- Most DFM concerns tacked on mold side



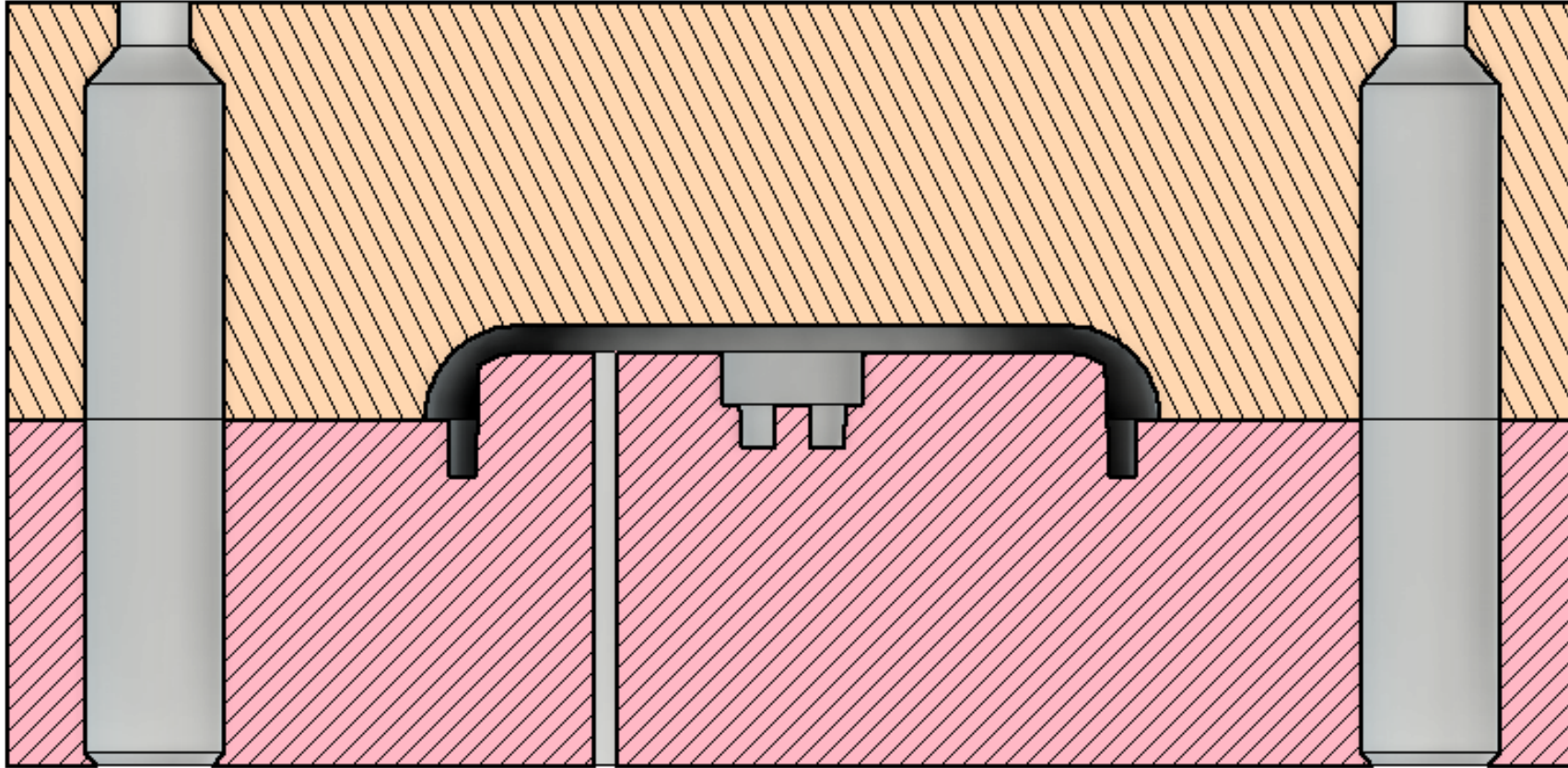
BOTTOM THERMOFORM



- Inner punch diam: 5/16"
- Outer punch diam: 2 3/8", same as top bubble
- Yoyo body and ring both leave room for burrs from punch process
 - 0.010" radially and vertically

MANUFACTURING

YOYO CUP MOLD



YOYO CUP CORE PROGRAM

Operation	Tool	Feature
Face	Face Mill	Core Top
Drill (Rapid Out)	Spot Drill	Ejector Pin Spots
Drill (Deep Drilling)	Ejector Pin Drill	Ejector Pin Holes
Adaptive	3/8" Flat EM	Core Roughing
Bore	3/8" Flat EM	Center bore halfway
Flat	3/8" Flat EM	Cavity-Core contact

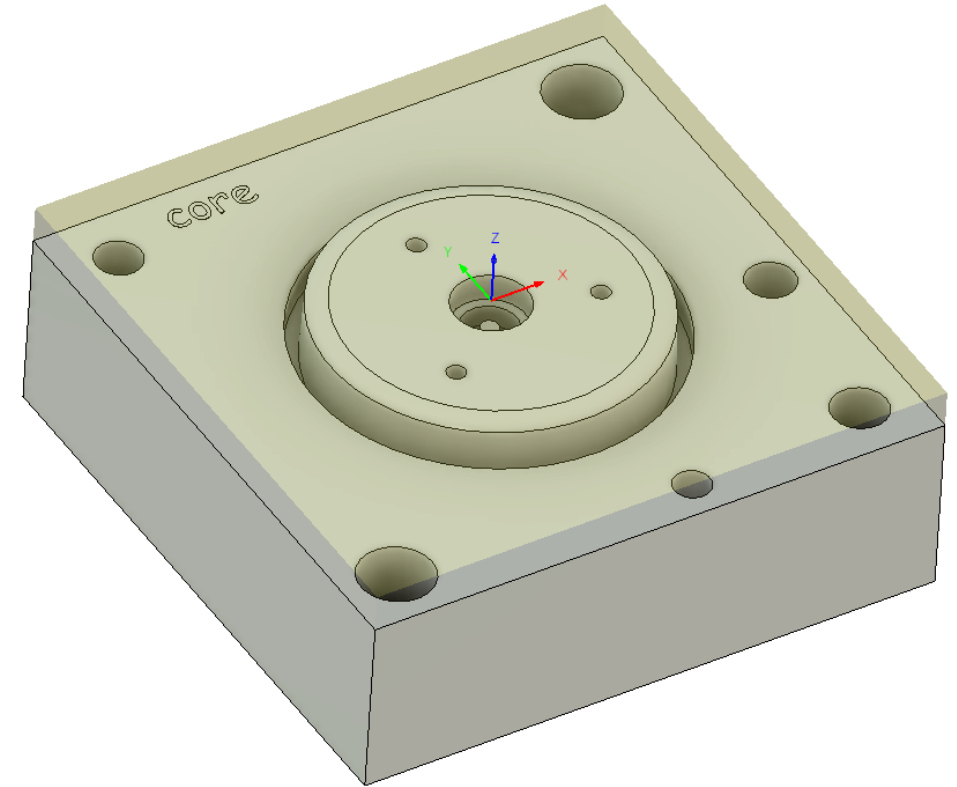
Run time: 50:53



YOYO CUP CORE PROGRAM CONT.

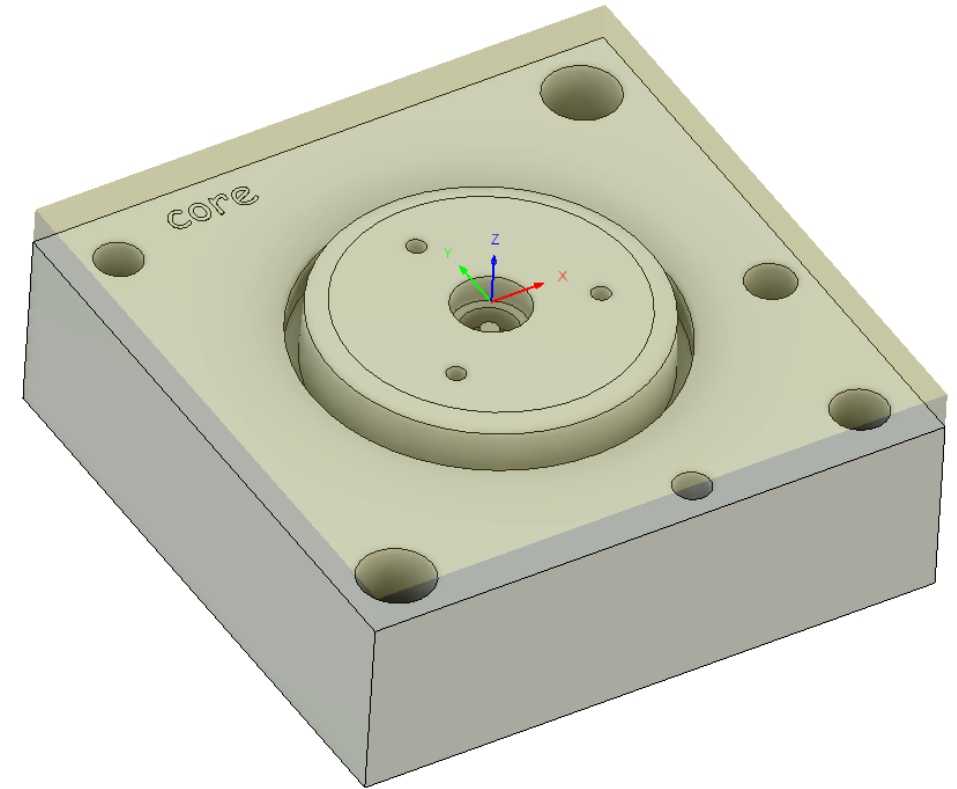
Operation	Tool	Feature
Bore	Taper EM	Core Taper (above contact face)
Contour	1/4" Ball EM	Yoyo Internal Radius
2D Contour	3/32" Flat EM	External Slot Roughing
Bore	Taper EM	Core Taper (below contact face)
2D Contour	3/32" Flat EM	Press Fit Surface Finishing
2D Contour	3/32" Flat EM	Center post roughing

Run time: 50:53



YOYO CUP CORE PROGRAM CONT.

Operation	Tool	Feature
Bore	Taper EM	Center Bore Internal Taper
Bore	3/32" Flat EM	Center Post Press Fit Finishing

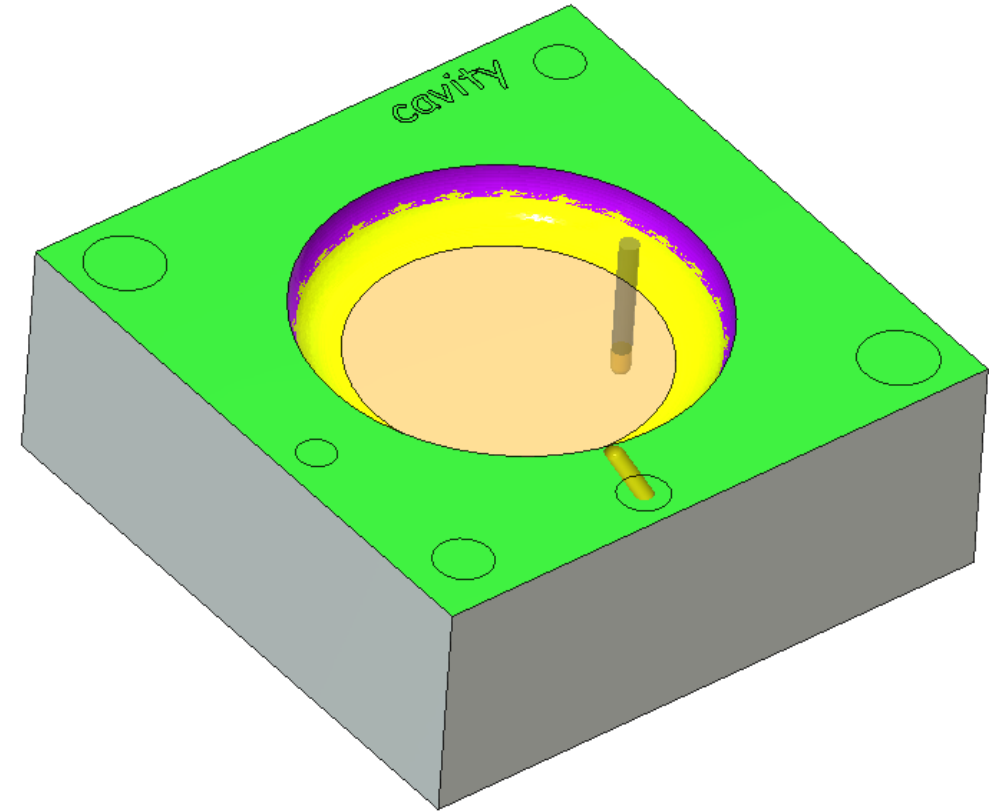


Run time: 50:53

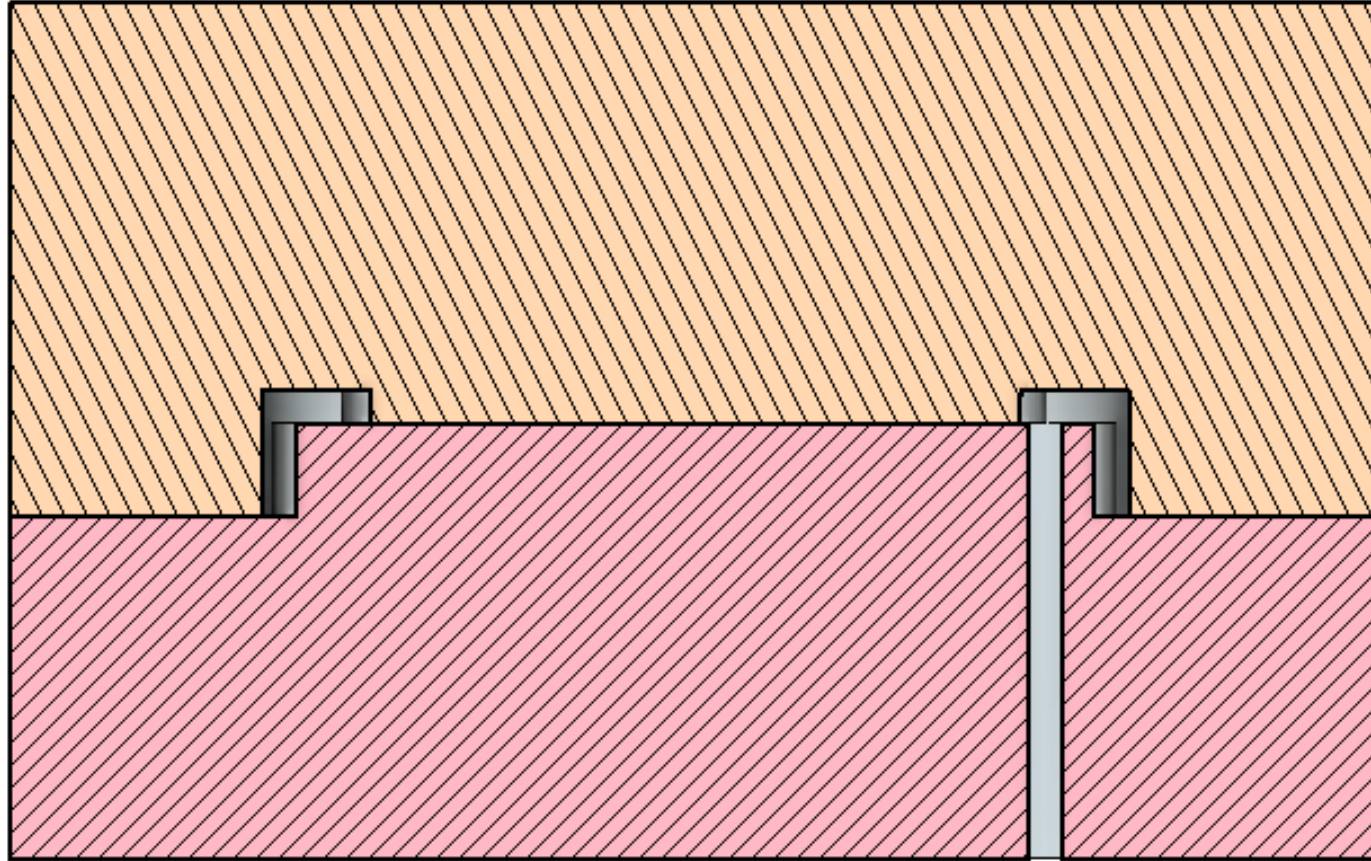
YOYO CUP CAVITY PROGRAM

Operation	Tool	Feature
Face	Face Mill	Cavity-Core Contact
Adaptive	3/8" Flat EM	Cavity Roughing
Flat	3/8" Flat EM	Flat bottom of cavity
Contour	1/4" Ball EM	Cavity Radius (higher stepdown)
Contour	1/4" Ball EM	Cavity Radius (lower stepdown)
2D Contour	1/8" Ball EM	Gate

Run time: 21:53

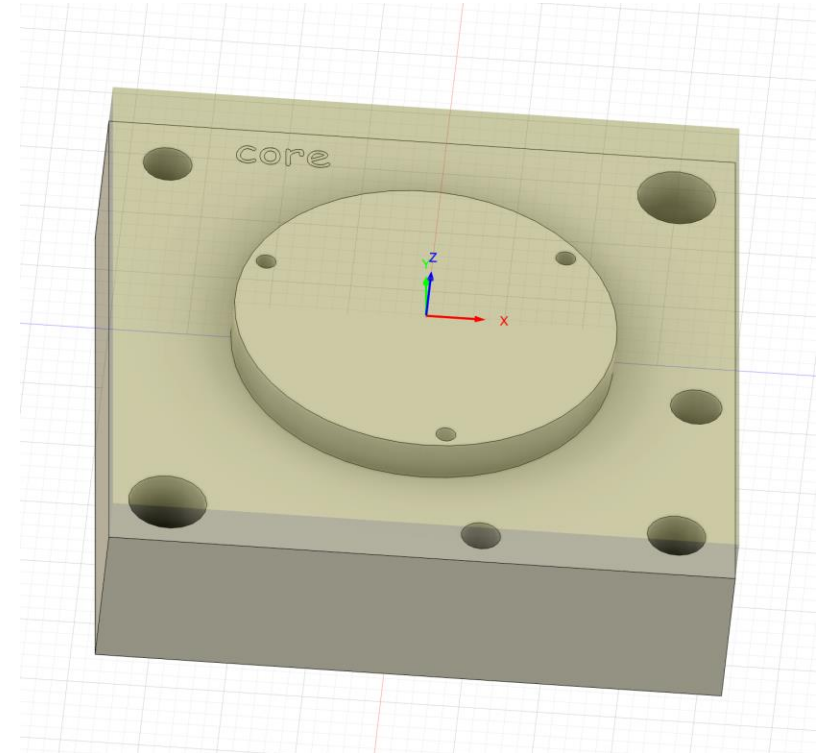


RING MOLD



RING CORE CAM PROGRAM

Operation	Tool	Feature
Face	Face Mill	Surface of mold
Spot Drill	1/8" 140° Spot Drill	Ejector pin guide holes
Drill ejector pin holes	0.116" Drill	Ejector pin holes
Adaptive	3/8" Flat endmill	Roughing pass for the core shape
Flat	3/8" Flat endmill	Finishing pass
2D Contour	3/8" Flat endmill	Press fit surface finishing pass

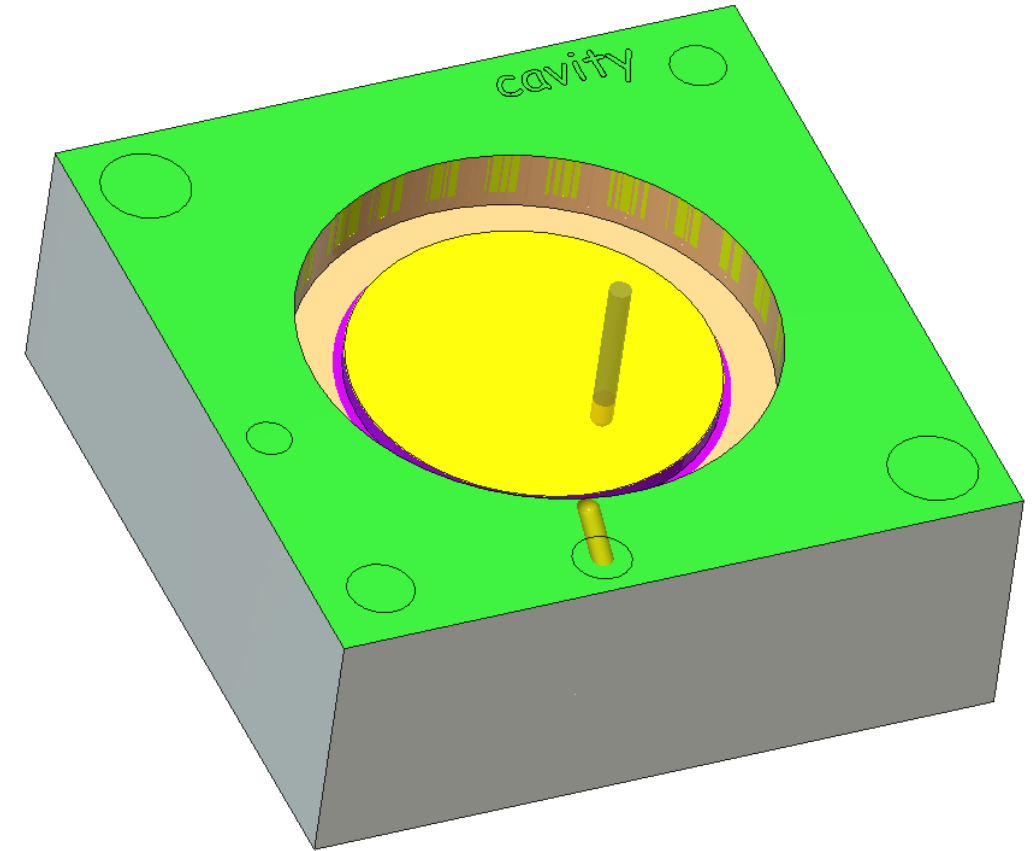


Run time: 15:33

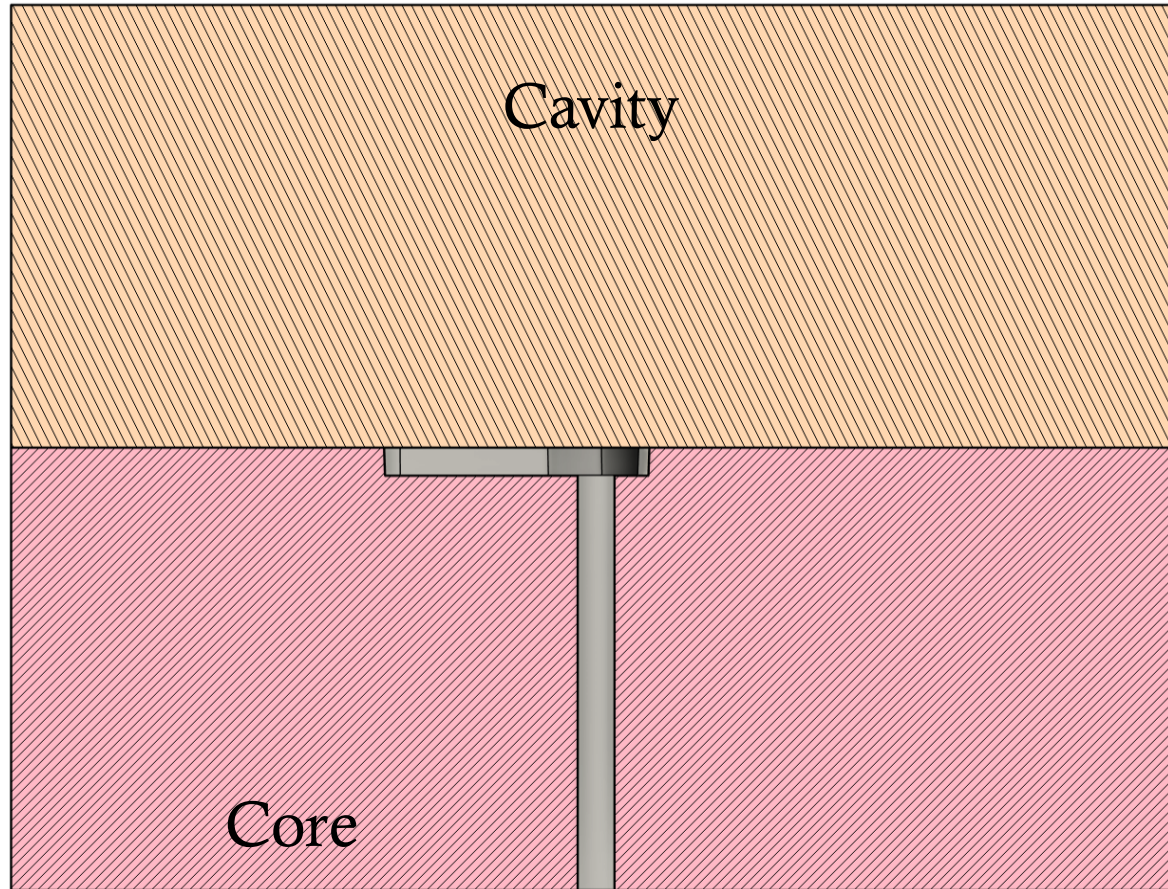
RING CAVITY CAM PROGRAM

Operation	Tool	Feature
Face	Face Mill	Surface of mold
Adaptive	3/8" Flat endmill	
2D Contour	1/4" Flat endmill	
2D Contour	1/4" Flat endmill	
Flat	3/8" Flat endmill	
2D Contour	1/8" Ball endmill	

Run time: 16:50



HOUR-MINUTE HAND MOLD

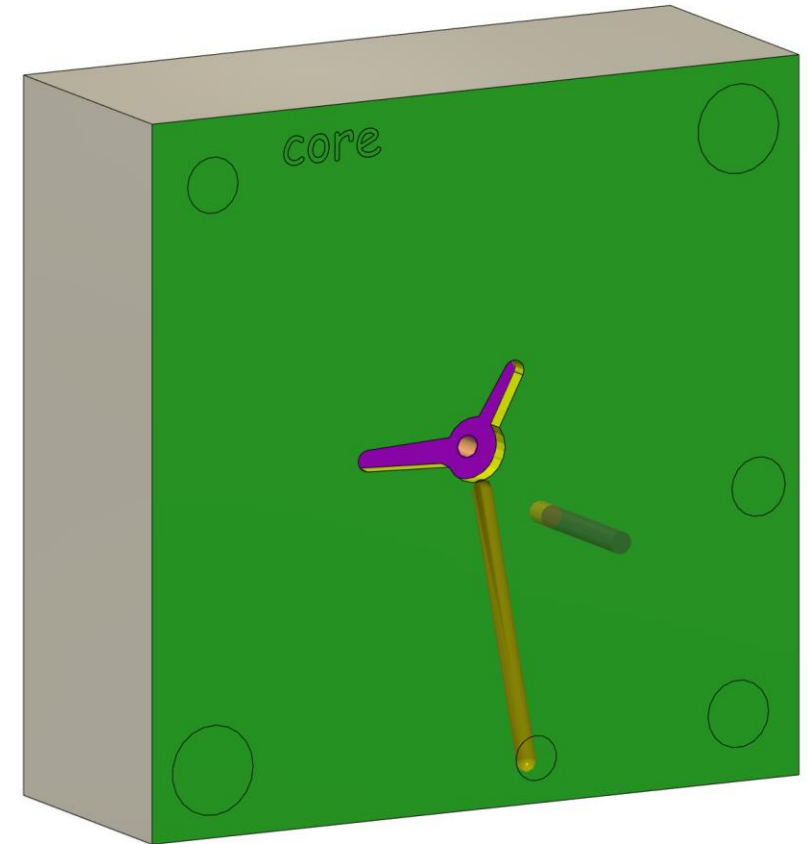


Using ejector pin as diameter
for the press fit pin

May need custom ejection
pin for correct press fit pin
length

HOUR-MINUTE HAND CORE PROGRAM

Operation	Tool	Feature
Face	Face Mill	Facing op
Spot drill	Spot Drill	Spot drill ejector pin hole
Peck drilling	Ejector Pin Drill	Ejector pin hole (also press fit pin hole)
2D Adaptive	3/32" Endmill	Hands cavity
Drafted sides	Taper Endmill	2 deg taper on sides
Gate	Ball Endmill	Gate

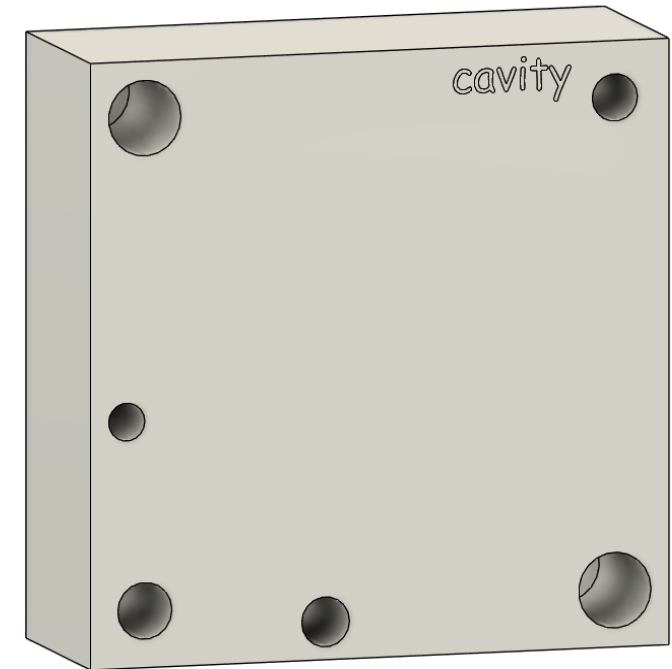


Run time: 15:28

HOUR-MINUTE HAND CAVITY PROGRAM

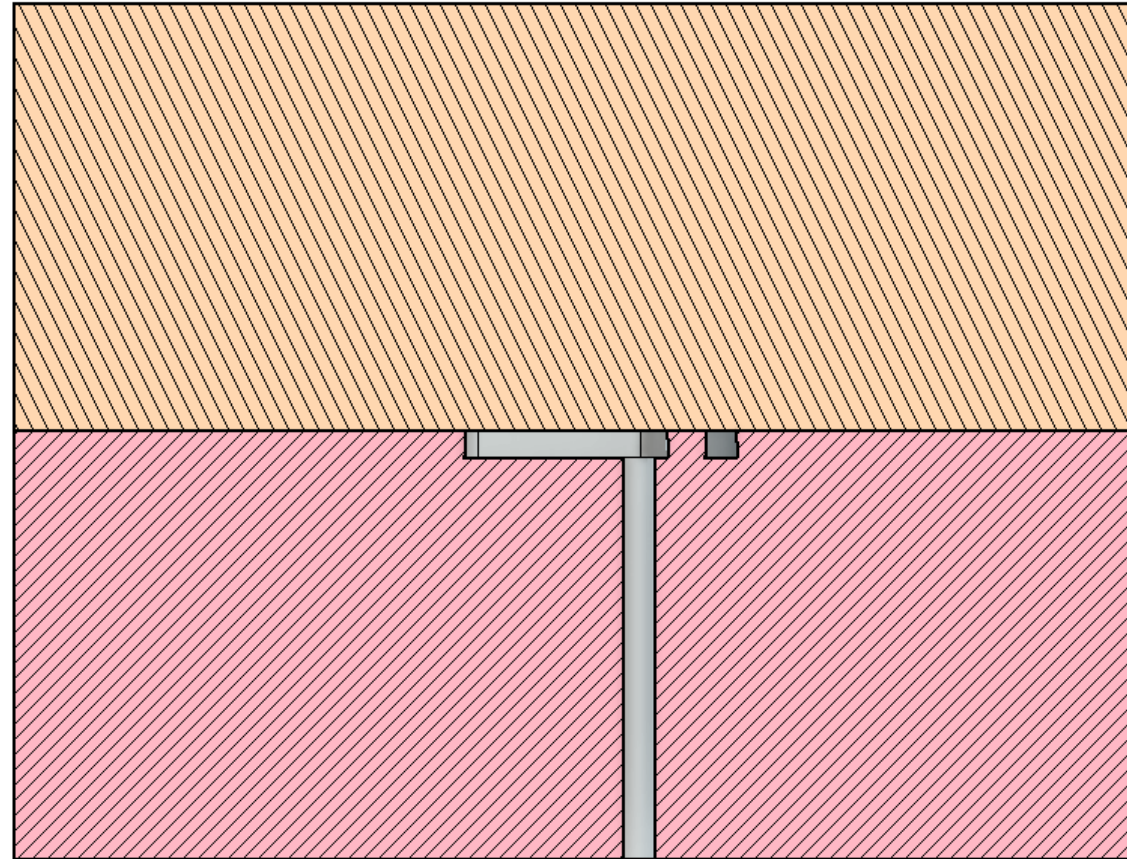
Operation	Tool	Feature
Face	Face Mill	Cavity-Core Contact

Same cavity for both hour/minute and seconds hand, this side can be used for both



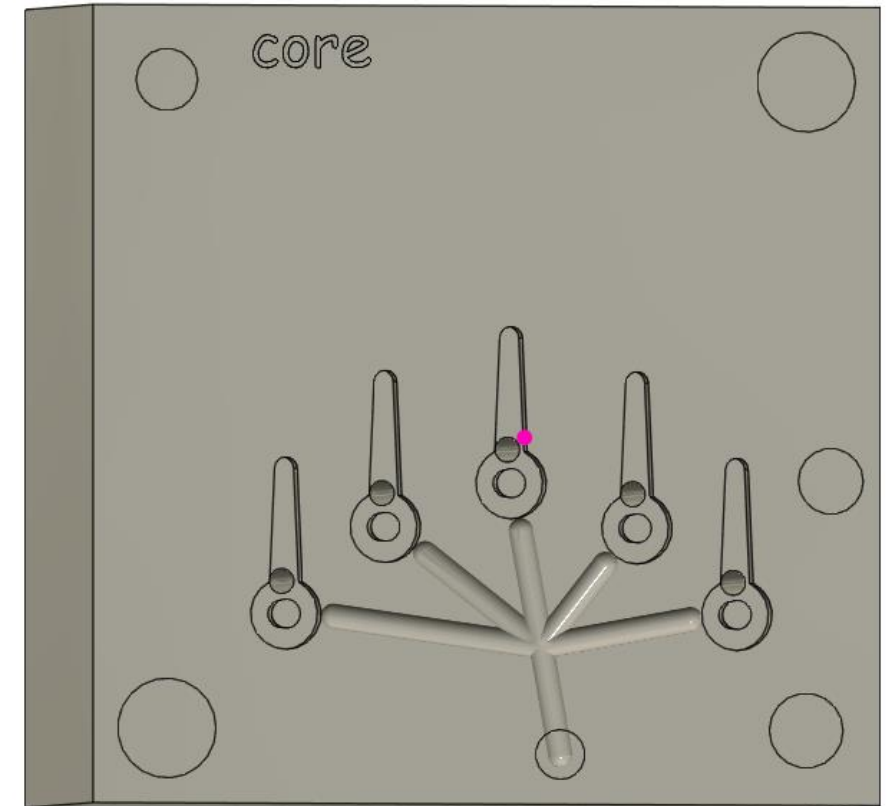
Run time: 1:09

SECOND HANDS MOLD



SECONDS HAND CORE PROGRAM

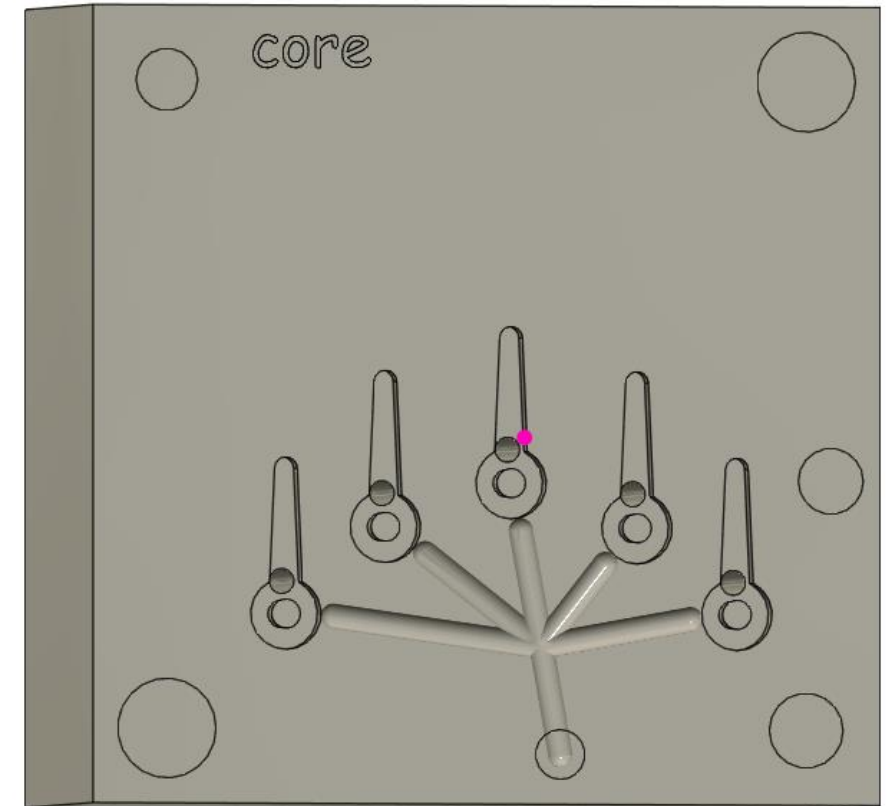
Operation	Tool	Feature
Face	Face Mill	Core Top
Drill (Rapid Drill)	Spot Drill	Ejector Pin Spots
Drill (Deep Drilling)	Ejector Pin Drill	Ejector Pin Holes
2D Contour	3/32" EM	Hands Cavity
2D Contour	1/8" Ball EM	Runners



Run time: 22:35

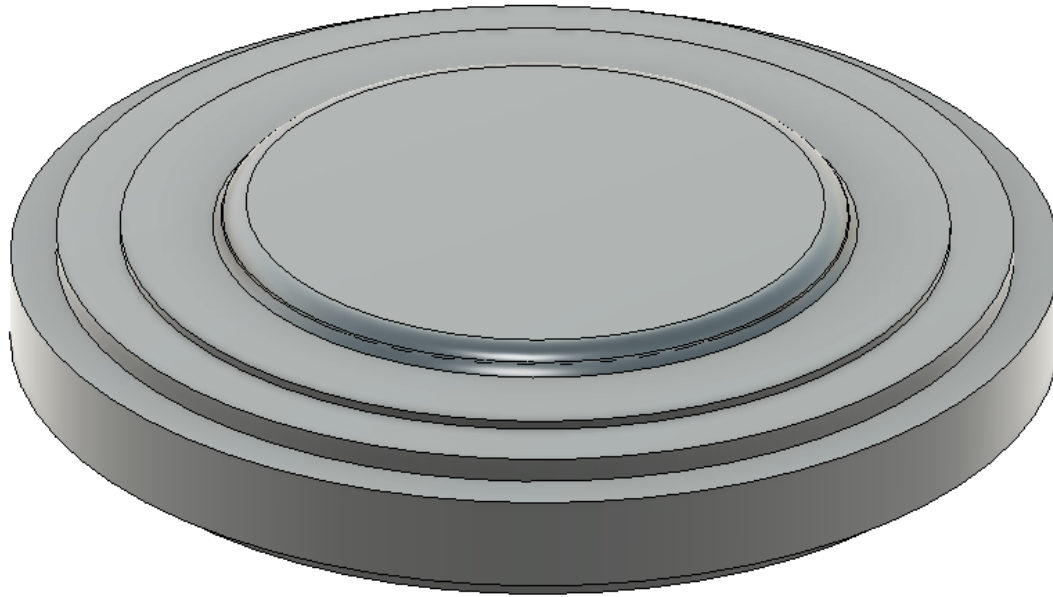
SECONDS HAND CORE PROGRAM CONT.

Operation	Tool	Feature
2D Contour	1/16" 2 Taper EM	Taper on Outside Edge
2D Contour	1/16" 2 Taper EM	Taper Around Inner Post



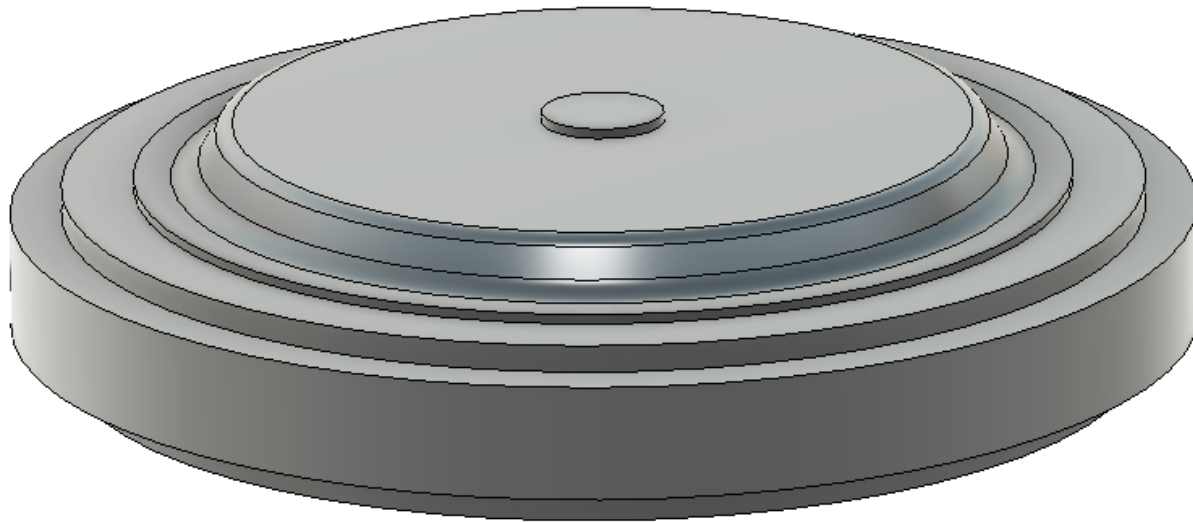
Run time: 22:35

TOP THERMOFORM MOLD



- Designed to interface with center hole of fixture bottom as well as fixture ring
 - Toleranced such that it fits nicely in fixture after burrs from supports sanded off
- Extra raised lip at punch diameter
 - Intended to make alignment easier by adding feature that lines up with punch
 - Currently existing mold has 0.01" tall lip, but increasing that to 0.025" next time (pictured)

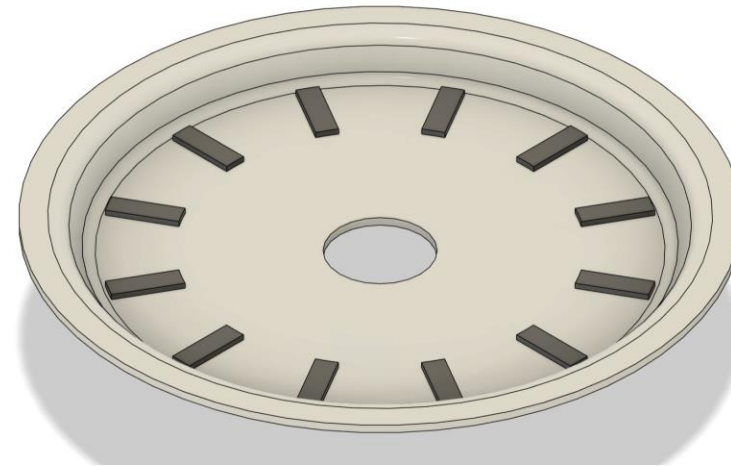
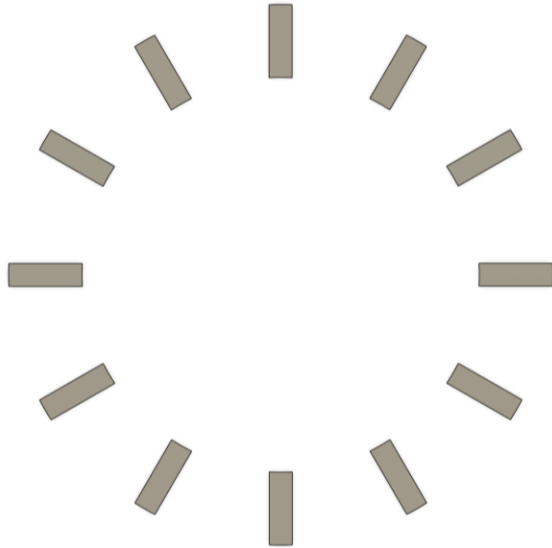
BOTTOM THERMOFORM MOLD



- Same base geometry, so will fit in fixture in same way as top mold
- Extra raised lip at both punch diameters
 - Intended to make alignment easier by adding feature that lines up with punch
 - Or line up the punch itself in case of center hole
- May add vacuum holes depending on results from hole-less mold (once it's printed)

STICKER

- Vinyl cut out shown in left image
 - Ticks on white donut shape, using punch one size up from thermoform
- Applied to bottom thermoform with jig
- Sample vinyl sticker to be cut to test application to create jig



PATH FORWARD

Week of	Objective	Risks
March 23rd	Finish machining all IM molds, all IM parts made at least once, all TF made at least once	All are first drafts, could be wrong geometry
March 30th	Update molds with learnings, have a fully assembled yoyo half	Could be a lot of learnings
April 5th	Add overmolded nuts	Could change how IM is working
April 13th	1 yoyo fully assembled	May need to tweak designs/specs
April 20th	Manufacturing Review, 25 yoyos fully assembled	Big ask/chunk of work to complete
April 27th	50 yoyos completed, our own drop test	Will have to remanufacture some if they fail
May 4th	All 50 yoyos assembled, pass the drop test!	

RISK AND MITIGATIONS

Part	Risk	Mitigation
Minute hand press fit	Press fit is not durable/doesn't work	Adjust TF geometry top bubble to hold it in, otherwise sticker
Two TF plates in between ring and body	Ring/body press fit is loosened	Adjust IM ring geometry or remove 1 TF part
Thermoformed clock face	Misalignment of the center hole	Make a jig to auto-align the part before punching a hole, or include a post in the mold that can later be more easily cut off to form a hole
	Running behind on schedule	Merge minute/hour hand with second hand as one part

THANK YOU!

APPENDIX

APPENDIX 1: PRESS FIT DIMENSION DERIVATION

Ring & Body:

$$p = \frac{E}{R} \delta \left(\frac{(r_o^2 - R^2)(R^2 - r_i^2)}{2R^2(r_o^2 - r_i^2)} \right) \rightarrow F_{insert} = \mu p 2\pi R x_{insert}$$

$$\delta = \frac{F}{2\pi\mu R x} \frac{1}{\frac{E}{R} \left(\frac{(r_o^2 - R^2)(R^2 - r_i^2)}{2R^2(r_o^2 - r_i^2)} \right)} = \frac{F R^2 (r_o^2 - r_i^2)}{\pi \mu x E (r_o^2 - R^2)(R^2 - r_i^2)}$$

$$R = 1.198 \text{ in}$$

$$r_o = R + 0.095$$

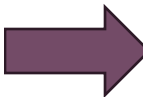
$$r_i = R - 0.095$$

$$F = 366 \text{ N}$$

$$x = 0.2 \text{ in}$$

$$E = 1.3 \text{ GPa}$$

$$\mu = 0.76$$


$$\delta = 0.012 \text{ in}$$

APPENDIX 1: PRESS FIT DIMENSION DERIVATION

Hour/Minute & Body:

$$p = \frac{E}{R} \delta \left(\frac{(r_o^2 - R^2)(R^2 - r_i^2)}{2R^2(r_o^2 - r_i^2)} \right) \rightarrow F_{insert} = \mu p 2\pi R x_{insert}$$

$$\delta = \frac{F}{2\pi \mu R x} \frac{1}{\frac{E}{R} \left(\frac{(r_o^2 - R^2)(R^2 - r_i^2)}{2R^2(r_o^2 - r_i^2)} \right)} = \frac{F R^2 (r_o^2 - r_i^2)}{\pi \mu x E (r_o^2 - R^2)(R^2 - r_i^2)}$$

$$R = 0.06375 \text{ in}$$

$$F = 366 \text{ N}$$

$$r_o = .1875 \text{ in}$$

$$r_i = 0 \text{ in}$$

$$\delta = 0.0015 \text{ in}$$

$$x = 0.142 \text{ in}$$

$$F = 500 \text{ N}$$

$$E = 1.3 \text{ GPa}$$

$$\mu = 0.76$$

$$\delta = 0.002 \text{ in}$$